DR. MAHALINGAM COLLEGE OF ENGINEERING AND TECHNOLOGY

Department of Career Planning and Guidance

Aptitude & Logical Reasoning Assessment (APT60)

CANDIDATE QUESTION PAPER

EXAMINATION COPY

Assessment Code: APT60
Duration: 60 Minutes
Total Questions: 60 MCQs
Marking: +1.0 Mark | -1.0 Negative Mark

Student Name: _____ **Roll Number:** _____ **Date:** _____

Question 1 [+1 Mark, -1 Negative Mark]

Three numbers are in the ratio of 3 : 4 : 5 and their L.C.M. is 2400. Their H.C.F. is:

A 40
B 80
C 120
D 200

Question 2 [+1 Mark, -1 Negative Mark]

The ratio of two numbers is 3 : 4 and their H.C.F. is 4. Their L.C.M. is:

A 12
B 16
C 24
D 48

Question 3 [+1 Mark, -1 Negative Mark]

The L.C.M. of two numbers is 48. The numbers are in the ratio 2 : 3. Then the sum of the numbers is:

A 28
B 32
C 40
D 64

Question 4 [+1 Mark, -1 Negative Mark]

If one-third of one-fourth of a number is 15, then three-tenth of that number is:

A 35
B 36
C 45
D 54

Question 5 [+1 Mark, -1 Negative Mark]

Three times the first of three consecutive odd integers is 3 more than twice the third. The third integer is:

A 9
B 11
C 13
D 15

Question 6 [+1 Mark, -1 Negative Mark]

The difference between a two-digit number and the number obtained by interchanging the positions of its digits is 36. What is the difference between the two digits of that number?

A 3
B 4
C 9
D Cannot be determined

Question 7 [+1 Mark, -1 Negative Mark]

A two-digit number is such that the product of the digits is 8. When 18 is added to the number, then the digits are reversed. The number is:

A 18
B 24
C 42
D 81

Question 8 [+1 Mark, -1 Negative Mark]

If the number 481 * 673 is completely divisible by 9, then the smallest whole number in place of * will be:

A 2
B 5
C 6
D 7

Question 9 [+1 Mark, -1 Negative Mark]

The value of a washing machine depreciates at the rate of 10% every year. If its present value is Rs. 8,748, then what was the price of the washing machine three years ago?

A Rs. 10,000
B Rs. 11,000
C Rs. 12,000
D Rs. 15,000

Question 10 [+1 Mark, -1 Negative Mark]

A student erroneously multiplied a number by 2/5 instead of 5/2. What is the percentage error in the calculation?

A 24%
B 54%
C 74%
D 84%

Question 11 [+1 Mark, -1 Negative Mark]

Ram spends 70% of his salary and deposits 15% of his salary in the bank. If he is left with Rs. 1,500, what is his monthly salary?

A Rs. 9,000
B Rs. 9,500
C Rs. 10,000
D Rs. 10,500

Question 12 [+1 Mark, -1 Negative Mark]

A fruit seller had some apples. He sells 40% apples and still has 420 apples. Originally, he had:

A 588 apples
B 600 apples
C 672 apples
D 700 apples

Question 13 [+1 Mark, -1 Negative Mark]

The price of an article was first decreased by 20% and then increased by 15%. What was the overall percent decrease or increase?

A 8% increase

B 8% decrease

C 2% decrease

D 2% increase

Question 14 [+1 Mark, -1 Negative Mark]

A train travelled a certain distance at a uniform speed. Had the speed been 6 km/h more, it would have needed 4 hours less. Had the speed been 6 km/h less, it would have needed 6 hours more. The distance, in km, travelled by the train is:

A 620 km

B 720 km

C 540 km

D 840 km

Question 15 [+1 Mark, -1 Negative Mark]

A boat's speed in still water is 22 km/h, while the river is flowing with a speed of 8 km/h. The time taken to cover a certain distance upstream is 4 hours more than the time taken to cover the same distance downstream. Find the distance.

A 100 km

B 105 km

C 110 km

D 115 km

Question 16 [+1 Mark, -1 Negative Mark]

A boat goes 20 km upstream and 44 km downstream in 8 hours. In 5 hours, it goes 15 km upstream and 22 km downstream. Determine the speed of the boat in still water.

A 5 km/h

B 8 km/h

C 10 km/h

D 12 km/h

Question 17 [+1 Mark, -1 Negative Mark]

A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is:

A 5/7

B 8/15

C 9/16

D 3/10

Question 18 [+1 Mark, -1 Negative Mark]

A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

A 5 days

B 10 days

C 15 days

D 20 days

Question 19 [+1 Mark, -1 Negative Mark]

A:B is 1:2; B:C is 3:2 and C:D is 1:3. Find A:B:C:D.

A 3:6:4:12

B 2:3:5:7

C 3:5:7:6

D 2:1:6:13

Question 20 [+1 Mark, -1 Negative Mark]

The ratio of two numbers is 11 : 12. If the sum of the numbers is 253, find the difference between the numbers.

A 10

B 11

C 12

D 21

Question 21 [+1 Mark, -1 Negative Mark]

A sum of money is to be distributed among A, B, C, D in the proportion of 5 : 2 : 4 : 3. If C gets Rs. 1,000 more than D, what is B's share?

A Rs. 500

B Rs. 1,500

C Rs. 2,000

D None of these

Question 22 [+1 Mark, -1 Negative Mark]

The average weight of A, B and C is 45 kg. If the average weight of A and B be 40 kg and that of B and C be 43 kg, then the weight of B is:

A 30 kg

B 31 kg

C 32 kg

D 33 kg

Question 23 [+1 Mark, -1 Negative Mark]

The average weight of 36 men is 50 kg. It was found later that the figure of 37 kg was misread as 73 kg. What is the correct average?

A 40 kg

B 45 kg

C 49 kg

D 50 kg

Question 24 [+1 Mark, -1 Negative Mark]

The average age of 30 students is 9 years. If the teacher is included the average becomes 10 years. What is the age of the teacher?

A 30 years

B 35 years

C 40 years

D 45 years

Question 25 [+1 Mark, -1 Negative Mark]

Five years ago, the ages of A and B were 1 : 2. At present the ratio is 3 : 4. Find the present age of A.

A 5 years
B 6 years
C 7.5 years
D 8 years

Question 26 [+1 Mark, -1 Negative Mark]

Present ages of Sam and Anand are in the ratio of 5 : 4 respectively. Three years hence, the ratio of their ages will become 11 : 9 respectively. What is Anand's present age in years?

A 15 years
B 24 years
C 30 years
D 32 years

Question 27 [+1 Mark, -1 Negative Mark]

A father said to his son, 'I was as old as you are at the present at the time of your birth'. If the father's age is 38 years now, the son's age five years back was:

A 12 years
B 14 years
C 15 years
D 20 years

Question 28 [+1 Mark, -1 Negative Mark]

A bank offers 5% compound interest calculated on half-yearly basis. A customer deposits Rs. 1,600 each on 1st January and 1st July of a year. At the end of the year, the amount he would have gained by way of interest is:

A Rs. 80
B Rs. 121
C Rs. 145
D Rs. 189

Question 29 [+1 Mark, -1 Negative Mark]

The difference between simple and compound interests compounded annually on a certain sum of money for 2 years at 4% per annum is Re. 1. The sum (in Rs.) is:

A Rs. 597
B Rs. 625
C Rs. 720
D Rs. 856

Question 30 [+1 Mark, -1 Negative Mark]

A card is drawn from a pack of 52 cards. The probability of getting a queen of club or a king of heart is:

A 1/26
B 2/35
C 2/67
D 3/65

Question 31 [+1 Mark, -1 Negative Mark]

From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?

A 756
B 748
C 789
D 800

Question 32 [+1 Mark, -1 Negative Mark]

In how many different ways can the letters of the word 'LEADING' be arranged in such a way that the vowels always come together?

A 625
B 720
C 840
D 960

Question 33 [+1 Mark, -1 Negative Mark]

Rajeev's age after 15 years will be 5 times his age 5 years back. What is the present age of Rajeev?

A 5 years
B 10 years
C 15 years
D 20 years

Question 34 [+1 Mark, -1 Negative Mark]

The ratio of father's age to son's age is 4 : 1. The product of their ages is 196. What will be the ratio of their ages after 5 years?

A 5:11
B 11:6
C 11:4
D 11:7

Question 35 [+1 Mark, -1 Negative Mark]

A and B invest in a business in the ratio 3 : 2. If 5% of the total profit goes to charity and A's share is Rs. 855, the total profit is:

A Rs. 1,425
B Rs. 1,500
C Rs. 1,537.50
D Rs. 1,576

Question 36 [+1 Mark, -1 Negative Mark]

A starts business with Rs. 3,500 and after 5 months, B joins with A as his partner. After a year, the profit is divided in the ratio 2 : 3. What is B's contribution in the capital?

A Rs. 7,500
B Rs. 8,000
C Rs. 8,500
D Rs. 9,000

Question 37 [+1 Mark, -1 Negative Mark]

In what ratio should two varieties of sugar of Rs. 18 per kg and Rs. 24 per kg be mixed together to get a mixture whose cost is Rs. 20 per kg?

A 2 : 1
B 3 : 1
C 4 : 1
D 5 : 1

Question 38 [+1 Mark, -1 Negative Mark]

Two vessels A and B contain spirit and water in the ratio 5 : 2 and 7 : 6 respectively. Find the ratio in which this mixture be mixed to obtain a new mixture in the vessel C containing spirit and water in the ratio 8 : 5?

A 5 : 7
B 6 : 7
C 7 : 9
D 7 : 3

Question 39 [+1 Mark, -1 Negative Mark]

A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 25%. The percentage of water in the mixture is:

A 10%
B 20%
C 30%
D 40%

Question 40 [+1 Mark, -1 Negative Mark]

In a 729 litre mixture of milk and water, the ratio of milk to water is 7 : 2. To get a new mixture containing milk and water in the ratio 7 : 3, the amount of water to be added is:

A 50 L
B 65 L
C 80 L
D 81 L

Question 41 [+1 Mark, -1 Negative Mark]

At what rate percent per annum will a sum of money double in 20 years?

A 1.2%
B 2%
C 4%
D 5%

Question 42 [+1 Mark, -1 Negative Mark]

A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years. The sum is:

A Rs. 650
B Rs. 690
C Rs. 698
D Rs. 700

Question 43 [+1 Mark, -1 Negative Mark]

A shopkeeper sells an article for Rs. 200 with a loss of 20%. Find the cost price of the article.

A Rs. 220
B Rs. 250
C Rs. 280
D Rs. 260

Question 44 [+1 Mark, -1 Negative Mark]

S.P. of 10 candles is the same as C.P. of 12 candles. Find the gain percent.

A 11%
B 15%
C 20%
D 25%

Question 45 [+1 Mark, -1 Negative Mark]

If a retailer marks up his price by 30% and then allows a 30% discount, what is his percentage profit or loss?

A 9% gain
B 8% gain
C 8% loss
D 9% loss

Question 46 [+1 Mark, -1 Negative Mark]

Look at this series: 2, 6, 18, 54, ... What number should come next?

A 98
B 162
C 195
D 324

Question 47 [+1 Mark, -1 Negative Mark]

Look at this series: 22, 21, 23, 22, 24, 23, ... What number should come next?

A 20
B 25
C 30
D 45

Question 48 [+1 Mark, -1 Negative Mark]

Find the next number in the series: 3, 8, 18, 38, 78, ?

A 148
B 156
C 158
D 160

Question 49 [+1 Mark, -1 Negative Mark]

One morning Udai and Vishal were talking to each other face to face at a crossing. If Vishal's shadow was exactly to the left of Udai, which direction was Udai facing?

A East
B West
C North
D South

Question 50 [+1 Mark, -1 Negative Mark]

From his house, Lokesh went 15 km to the North. Then he turned West and covered 10 km. Then he turned South and covered 5 km. Finally turning to the East, he covered 10 km. In which direction is he from his house?

A East
B West
C North
D South

Question 51 [+1 Mark, -1 Negative Mark]

Rasik walked 20 m towards North. Then he turned right and walks 30 m. Then he turns right and walks 35 m. Then he turns left and walks 15 m. Finally he turns left and walks 15 m. In which direction and how many metres is he from the starting position?

A 15 m West
B 30 m East
C 30 m West
D 45 m East

Question 52 [+1 Mark, -1 Negative Mark]

Pointing to a photograph of a boy Suresh said, 'He is the son of the only son of my mother.' How is Suresh related to that boy?

A Brother
B Father
C Uncle
D Cousin

Question 53 [+1 Mark, -1 Negative Mark]

M is the son of N. N is the sister of O. O is the father of P. How is M related to P?

A Brother
B Cousin
C Nephew
D Uncle

Question 54 [+1 Mark, -1 Negative Mark]

Statements: All the windows are doors. No door is a wall.
Conclusions:
(I) Some windows are walls.
(II) No wall is a door.

A Only I follows
B Only II follows
C None follows
D Both follow

Question 55 [+1 Mark, -1 Negative Mark]

Statements: All cups are books. All books are shirts.
Conclusions:
(I) Some cups are not shirts.
(II) Some shirts are cups.

A Only I follows
B Only II follows
C None follows
D Both follow

Question 56 [+1 Mark, -1 Negative Mark]

Statements: No vehicle is a bicycle. Some bicycles are electric. All electric things require power.
Conclusions:
(I) Some things requiring power are bicycles.
(II) No vehicle requires power.

A Only conclusion I follows
B Only conclusion II follows
C Both conclusions follow
D Neither conclusion follows

Question 57 [+1 Mark, -1 Negative Mark]

P, Q, R, S, T, U and V are seven friends sitting in a circle facing the center. V is second to the left of S and is the neighbor of T and U. S is not a neighbor of R or T. P is a neighbor of Q and R. Which of the following is correct?

A Q is between P and S
B S is between U and P
C T is to the immediate right of V
D U is to the immediate left of V

Question 58 [+1 Mark, -1 Negative Mark]

Six files A, B, C, D, E and F are arranged by size. A is larger than C but smaller than E. B is smaller than D but larger than F. C is larger than D. Which file is the largest?

A A
B B
C E
D D

Question 59 [+1 Mark, -1 Negative Mark]

If DELHI is represented by its alphabetical positions as 4-5-12-8-9, how will CHENNAI be represented?

A 3-8-5-14-14-1-9
B 3-8-5-13-13-1-9
C 3-7-5-14-14-1-9
D 2-8-5-14-14-1-9

Question 60 [+1 Mark, -1 Negative Mark]

The HCF of two numbers is 18 and their LCM is 1260. If one number is 90, what is the other number?

A 216
B 252
C 270
D 280